

CS 198: Quantum Algorithms Decal Syllabus

Course Information

Title: Quantum Algorithms

Unit Value: 2

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Lecture Time and Location: –

Office Hours Time and Location: – @ Campbell 101

Course Description

Prerequisites

- Linear Algebra (Math 54, Physics 89 or equivalent) and python programming experience (e.g. CS 61A or equivalent). Knowledge of Physics 191A, or concurrent enrollment, is *strongly preferred*. No instructor permission required.

Overview

- The purpose of this course is to survey the development of quantum algorithms, from Shor's original breakthrough to more recent approaches. Emphasis is placed on hands-on experience, including implementation using Qiskit (AER).
- Students will learn about extensions on classic results, gaining a modern perspective on original algorithms. Special topics lectures inspire students to explore their own interests.
- The primary purpose is to inspire more students to participate in quantum computing by exploring its current research and inspecting its utility. A final project will serve as a capstone of their achievement.

Learning Objectives

- Be very comfortable with well-known algorithms such as Grover, Shor, Quantum Phase Estimation.
- Understand the basics of algorithmic analysis applied to quantum algorithms, including results on quantum advantage.
- Hands-on experience with implementing quantum algorithms using Qiskit.
- Learn about algorithms developed since, developing a well thought out final project that involves one of these approaches.

Methods of Instruction

- Lectures (two 1h lectures per week)

- Final Project

Materials

Primary books/reading for the course

- Students are required to read the sections corresponding to lecture for the texts below:
 - <https://homepages.cwi.nl/~rdewolf/qcnotes.pdf> (de Wolf)
 - https://math.berkeley.edu/~linlin/qasc/live_notes_0429.pdf (Lin)
 - <https://www.cs.umd.edu/~amchilds/qa/qa.pdf> (Childs)

Supplemental materials for the course

- Students are encouraged to learn more by going over the following materials
 - Nielsen, Michael A.; Chuang, Isaac L. (2010). *Quantum Computing and Quantum Information*. Cambridge University Press.
 - Kitaev, A. Y., Shen, A. H., & Vyalyi, M. N. (2002). *Classical and quantum computation*. American Mathematical Society.
 - Quantum Algorithm Zoo <https://quantumalgorithmzoo.org/#ONML>
 - Qiskit Textbook: <https://qiskit.org/textbook/preface.html>

Website

- Our website <https://qcb.studentorg.berkeley.edu/> contains additional resources and examples of past projects.

Requirements

Lectures: Attending the two lectures per week

Homeworks: Optional. Lab notebooks will be provided for you to implement in Qiskit.

Final Project: A capstone project on a topic of your choice (due __)

- The point of a project is for you to explore an algorithm not discussed in class (or an extension of what we talked about), and present it to the class.
- Implementation of your algorithm of choice is required.
- Submitting a final project is *required* to pass the course.

Policies

Grading: This course is graded on a P/NP basis. Regular attendance, participation and completion of assignments is required for a Pass grade. We have 2 unexcused absences. Excused absences are allowed, but students will need to email the instructors in advance, and after lecture slides are posted. Lectures will not be recorded. A final project must be completed in order to pass.

Attendance 20%

Final Project 80%

Schedule

Week	Lecture 1	Lecture 2
UNIT 1 Introductory Material		
1	Logistics, Qiskit Intro, Quantum Circuits <i>Required Readings: de Wolf ch. 2.1-3</i>	Qiskit and Quantum Circuits <i>Required Readings: n/a</i>
2	Deutsch-Jozsa Algorithm <i>Required Readings: de Wolf ch. 2.4</i>	Quantum Complexity (1) <i>Required Readings: Childs ch 2.3</i>
3	Quantum Complexity (1) <i>Required Readings: de Wolf ch. 13</i>	Quantum Complexity (2) <i>Required Readings: de Wolf ch. 14</i>
UNIT 2 Algorithms		
4	Quantum Fourier Transform and Quantum Phase Estimation <i>Required Readings: Lin ch. 16.1, 16.3</i>	Grover's Algorithm and Quantum Counting <i>Required Readings: Lin 11.1</i>
5	Shor's Algorithm <i>Required Readings: de Wolf ch. 5</i>	Adiabatic Quantum Computation <i>Required Readings: Childs ch. 31</i>
6	Variational Approaches (1) <i>Required Readings: n/a</i>	Variational Approaches (2) <i>Required Readings: n/a</i>
UNIT 3 Hamiltonian Simulations and Beyond		
7	Block Encoding <i>Required Readings: Lin ch. 9</i>	Qubitization <i>Required Readings: Lin ch. 10</i>
8	Quantum Signal Processing <i>Required Readings: Lin ch. 12</i>	Quantum Singular Value Transformation (1) <i>Required Readings: Lin ch. 13.2-3</i>
9	Quantum Singular Value Transformation (2) <i>Required Readings: Lin ch. 13</i>	Block Encoding based Hamiltonian Simulation <i>Required Readings: Lin ch. 14.1-2</i>
10	Operator Splitting based Hamiltonian Simulation <i>Required Readings: Lin 15.1-3</i>	Quantum Walks (1) <i>Required Readings: Lin ch. 17</i>
11	Quantum Walks (2) <i>Required Readings: Lin ch. 17</i>	HHL Algorithm <i>Required Readings: de Wolf ch. 10</i>
UNIT 4 Special Topics (Open Quantum Systems? AI?)		
12	TBD	TBD

13	Final Project Workshop	Final Project Workshop
UNIT 5 Final Project		
Final Project Presentations		

Resources

Office Hours

- Held every week

Tips for success

- Be engaged in discussions.
- Be curious about the content.
- Be brave to dive deep.

Accessibility and Accommodations

All students who have special needs can receive appropriate accommodations. The DSP office (<http://www.dsp.berkeley.edu>) must determine or verify these accommodations before they can be offered. Students who are requesting academic accommodations are responsible for contacting the DSP Coordinator immediately. Please contact the instructor when a request for accommodation has been filed.

Equity and Inclusion

In accordance with applicable Federal and State law and University policy, the University of California, Berkeley, does not discriminate on the basis of race, color, national origin, religion, sex, gender identity, pregnancy, physical or mental disability, medical condition, ancestry, marital status, age, sexual orientation, citizenship, or service in the uniformed services. The University of California, Berkeley also prohibits sexual harassment.

We affirm the dignity of all people and promote diversity, equity, inclusion and belonging in all that we do. We acknowledge that biases and injustices (both historic and current) permeate all aspects of our society and we strive to ensure that every person with whom we interact feels supported, valued and respected.

We encourage you to approach your instructor or Student Services if: official records do not reflect your correct name and/or set of pronouns that you would like us to use; your performance in the class is impacted by your experiences outside the class (e.g., family matters, current events); we would like to help you find resources to cope; something was said in class (by anyone) that made you feel uncomfortable.

Evaluation of the course

Feedback: Each week we will have an attendance form, in which students can voice anything that they like and anything that they wish were different

End-of-course evaluation: There will be an anonymous end-of-course survey

Disclaimer

The syllabus is subject to change.